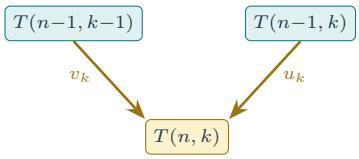
COMBINATORICS ON ONE PAGE

The general sequencing law for triangles

One bidiagonal rule with two sets of weights produces the Pascal, Stirling, Worpitzky and Eulerian triangles.

1 / THE RULE AND THE PICTURE

Each entry is a weighted sum of the two entries above it



$$\boxed{T(n,k) = u_k\,T(n-1,k) + v_k\,T(n-1,k-1)}$$

Stacking a row as a column vector, one step of the law is multiplication by a lower bidiagonal matrix:

$$\begin{pmatrix} T(n,0) \\ T(n,1) \\ T(n,2) \end{pmatrix} = \begin{pmatrix} u_0 & & \\ v_1 & u_1 & \\ & v_2 & u_2 \end{pmatrix} \begin{pmatrix} T(n-1,0) \\ T(n-1,1) \\ T(n-1,2) \end{pmatrix}.$$

The weights are the only choices, and a triangular matrix has its diagonal u_k as its eigenvalues.

2 / FOUR TRIANGLES / THE SAME LAW, DIFFERENT WEIGHTS

Pascal, Stirling and Worpitzky with fixed weights, and Eulerian with weights that depend on the row

	Pascal $\binom{n}{k}$ weights $(1,1)$	**Stirling** $S(n,k)$ weights $(k,1)$	**Worpitzky** $W(n,k)$ weights (k,k)	**Eulerian** $A(n,k)$ weights $(k+1,\,n-k)$
$n=0$	1	1	1	1
$n=1$	1 1	1 1	1 2	1 1
$n=2$	1 2 1	1 3 1	1 6 6	1 4 1
$n=3$	1 3 3 1	1 7 6 1	1 14 36 24	1 11 11 1
$n=4$	1 4 6 4 1	1 15 25 10 1	1 30 150 240 120	1 26 66 26 1
$n=5$	1 5 10 10 5 1	1 31 90 65 15 1	1 62 540 1560 1800 720	1 57 302 302 57 1
$n=6$	1 6 15 20 15 6 1	$25 = 3\cdot 6 + 7$	$36 = 3(6+6)$	$66 = 3\cdot 11 + 3\cdot 11$
	$15 = 10 + 5$			

3 / WHAT EACH TRIANGLE COUNTS / AND ITS ROW SUMS

Subsets, set partitions, surjections and permutations by descents

triangle	counts	row sums	closed form
Pascal $\binom{n}{k}$	k-subsets of an n-set	$1, 2, 4, 8, 16, 32, 64$, that is 2^n	$\dfrac{n!}{k!\,(n-k)!}$
Stirling $S(n,k)$	partitions of n things into k blocks	Bell $1, 2, 5, 15, 52, 203$	$\dfrac{1}{k!}\sum_j (-1)^{k-j}\binom{k}{j}\,j^n$
Worpitzky $W(n,k)$	surjections from n things onto k labelled boxes	Fubini $1, 3, 13, 75, 541, 4683$	$\sum_j (-1)^{k-j}\binom{k}{j}\,j^n$
Eulerian $A(n,k)$	permutations of n with k descents	$n!$: $1, 2, 6, 24, 120, 720$	$\sum_j (-1)^j\binom{n+1}{j}(k+1-j)^n$

Labelling the blocks of a partition in $k!$ ways turns Stirling into Worpitzky, so $W(n,k) = k!\,S(n,k)$, and for example $W(4,2) = 2^4 - 2 = 14$ counts the maps from four things onto two boxes.

4 / HOW THE TRIANGLES LINK / GAUGE, BASIS AND THE BELL RECURRENCE

Pascal feeds Stirling through the weights, Worpitzky through a gauge, and Bell through the binomials

Replacing the Pascal weights $(1,1)$ by $(k,1)$ gives Stirling. Scaling row k by $k!$, the diagonal gauge $D = \mathrm{diag}(k!)$, sends v_k to $k\,v_k$ and carries Stirling to Worpitzky, while the diagonal $u_k = k$ is untouched, so both have the eigenvalues $1, \ldots, n$ and the characteristic polynomial $(\lambda-1)(\lambda-2)\cdots(\lambda-n)$. In the language of bases,

$$x^n = \sum_k S(n,k)\,x^{\underline{k}} = \sum_k W(n,k)\binom{x}{k}, \qquad B_{n+1} = \sum_k \binom{n}{k} B_k,$$

so the Worpitzky numbers are the coefficients of a power in Pascal's own basis, the falling factorial $x^{\underline{k}} = k!\binom{x}{k}$ is the change of basis between them, and the Bell numbers $B_n = \sum_k S(n,k)$ are built from binomials: $B_4 = 1 + 3\cdot 1 + 3\cdot 2 + 5 = 15$. The Eulerian numbers expand the same power in shifted binomials, $x^n = \sum_k A(n,k)\binom{x+k}{n}$, which is Worpitzky's identity proper.

ASIDE / WEIGHTS THAT DEPEND ON n

The Eulerian weights $(k+1,\,n-k)$ depend on the row, as do those of the unsigned Stirling numbers of the first kind, $(n-1,1)$, whose row sums are also $n!$.

LINKS TO OTHER SHEETS

The characteristic equation and the gauge: *The characteristic equation beneath five presentations*; the determinant as a product of eigenvalues: *The determinant in its many guises*.

Convention. $T(n,k)$ is the entry in row n, column k; $x^{\underline{k}} = x(x-1)\cdots(x-k+1)$; S and W are indexed from $k=1$.

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